Finite-blocklength Linear Coding with Strictly Causal Interference Side Information at Encoder

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***Abstract*—Channels with strictly causal interference side information arise when the encoder has access to the interference sequence only up to the channel use $i-1$ at the channel use i. We refer to this class of channels as strictly-causally known interference channels. Despite their prevalence in sequential communication systems, finite-blocklength code design for such channels remain largely unexplored. To address the challenge, this paper proposes a coding framework formulated from a feedforward control perspective. Within this framework, a linear coding scheme is derived by formulating and solving a relative sensitivity minimization problem. Finite-blocklength analysis indicates that for binary messages and any blocklength greater than one, positive interference power gives a strict reliability improvement over the optimal no-side-information code. For a fixed target error probability and sufficiently large but fixed blocklength, a rate improvement is established when the interference-to-noise ratio is sufficiently large.**

***Index Terms*—Strictly-causally known side information, feedforward control, linear coding, finite blocklength analysis.**

I. INTRODUCTION

Encoder state information is non-causal, causal, or strictly causal when it reveals the entire state sequence, the current and past states, or only the past states, respectively. Remarkable progress has been made in channel coding for non-causally and causally known interference. Prior studies show that noncausal [1] and causal [2] knowledge of interference side information at the encoder can increase channel capacity compared with the corresponding channels without access to side information at the encoder. In contrast, the strictly-causal setting is far more constrained. It is shown in [3] that when the side information is independent and identically distributed (i.i.d.), access to strictly causal side information does not increase the channel capacity. Modern channel coding techniques, such as low-density parity-check (LDPC) [4], polar [5], [6], and turbo [7] codes approach the channel capacity without side information, but their performance exhibits a substantial gap from capacity at finite blocklengths [8]. To the best of our knowledge, finite-blocklength code design for strictly-causally known interference channels remains an open problem and requires fundamentally different design principles.

To address this challenge, this paper develops a coding framework from a feedforward control perspective [9], as both strictly-casually known interference channel and feedforward control system operate in an *open-loop* manner with access to partial side information [10]. Specially, the proposed coding framework transforms the encoder and decoder designs into two iterative learning control (ILC) synthesis problems [11]. ILC is a feedforward algorithm that exploits the strictly causal system knowledge to handle repetitive tasks subject to system *disturbances* (all channel impairments including interference and noise). Its stability and convergence has been extensively studied [12]–[14] over various system conditions. By leveraging historical trial-and-error experience, ILC [15] progressively uses a *learning function* and a *robust filter* to update the feedforward input such that the system output is driven toward the desired reference signal.

For the realization over the strictly-casually known interference channel, successive rehearsal trials are constructed from past side information. The encoder iteratively accumulates the rehearsal errors to update subsequent transmissions, while the decoder iteratively estimates a fixed message symbol based on the successively received signals, which is equivalent to an ILC process with measurement disturbances [16], [17].

Within this coding framework, a linear coding scheme is derived by formulating and solving a relative sensitivity minimization problem [18]. The analysis has two parts. It is shown for a message size of $M = 2$ and arbitrary blocklength N, the proposed linear scheme reduces the error probability ϵ to a range that is unattainable by a coding scheme without access to side information. For sufficiently large blocklength N and $M \geq 2$, our analysis use the finite-blocklength analysis framework from [8] to reveal that the scheme is advantageous in interference-dominated regimes. Specifically, when the interference power exceeds the noise power, we demonstrate the existence of a feasible target error probability ϵ for which the proposed linear scheme achieves a rate $R^\star > R^\star_{\text{no-SI},\epsilon}$, where $R^\star_{\text{no-SI},\epsilon}$ is the maximum rate achievable for error probability ϵ without side information. The full version of this paper is available at [19].

Notation. Scalars, vectors, and matrices are denoted, respectively, by lowercase, bold lowercase, and bold uppercase letters. x_a^b represents a sequence $x_a^b = \{x_a, x_{a+1}, \cdots, x_b\}$, for $b > a$. $\|\cdot\|_p$ denotes the p-norm. $|\cdot|$ denotes the

absolute value. The operators $\mathrm{tr}(\cdot)$ and $\det(\cdot)$ denote the matrix trace and determinant, respectively. $\mathcal{N}(\mu, \sigma^2)$ denotes a scalar Gaussian distribution with mean μ and variance σ^2. For a multivariate Gaussian distribution, $\mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$ denotes a Gaussian distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\mathbf{K}$. The notation $\overset{\text{i.i.d.}}{\sim}$ indicates that the random variables are independent and identically distributed. We write $f(x) = o(g(x))$ if $\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0$, and $f(x) = O(g(x))$ if $\limsup_{x\to\infty} \frac{|f(x)|}{g(x)} < \infty$.

II. System Model and Problem Statement

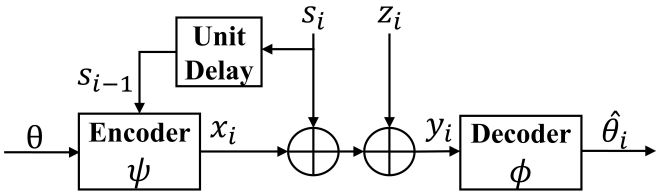


Fig. 1: System model of a strictly-casually known interference channel, where the encoder ψ generates the channel input x_i from θ and the past side information up to s_1^{i-1}, while the decoder ϕ estimates θ as $\hat{\theta}_i$ from the received signal y_1^i.

The system model of a strictly-casually known interference channel is presented in Fig. 1. The message symbol θ is uniformly drawn over a rate R (bits/channel use) message set $\mathcal{C} = \{\theta_1, \cdots, \theta_M\}$ with $\mathbb{E}[\theta^2] = 1$ and the message set size $M = 2^{NR}$, where the blocklength N is utilized for the transmission of $\theta \in \mathcal{C}$. The ith channel output is

$$y_i = x_i + s_i + z_i, \quad i = 1, \cdots, N, \tag{1}$$

where the interference $s_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_s^2)$ and additive noise $z_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ are mutually independent. The goal of this paper is to design a coding scheme for the channel model in Fig. 1, which consists of:

- An encoder generates $x_i = \psi_i(\theta, s_1^{i-1})$, $i = 1, \ldots, N$, with access to the interference side information $(s_1, \cdots, s_{i-1})$, that satisfies $\sum_{i=1}^{N} \mathbb{E}[x_i^2] \le NP$, where P is the average transmit power per channel use.
- A decoder sequentially decodes θ as $\hat{\theta}_i = \phi_i(y_1^i)$ and outputs $\hat{\theta} = \hat{\theta}_N$.

The performance of the proposed coding scheme is evaluated in terms of its finite-blocklength achievable rate $R^\star$. For a fixed blocklength N and a target error probability $\epsilon \in (0, 1)$, a rate $R^\star$ is achievable if the scheme supports $M \ge 2^{NR^\star}$ messages while satisfying the the maximal error constraint $P_{e,\max} = \max_{\theta\in\mathcal{C}} \Pr(\hat{\theta} \ne \theta \mid \theta) \le \epsilon$, where ϵ is the target probability of error. According to [3], access to strictly causal side information at the encoder does not increase channel capacity. In finite-blocklength regimes, we therefore define the baseline rate $R^\star_{\text{no-SI},\epsilon}$ as the maximum achievable rate with maximal error rate at most ϵ, over an additive white Gaussian noise (AWGN) interference channel without side information at the encoder in (1), where the effective channel signal-to-noise ratio (SNR) is given by $\rho = \frac{P}{\sigma_s^2+\sigma^2}$. For a given target probability of errorϵ, if the scheme's achievable rate satisfies $R^\star < R^\star_{\text{no-SI},\epsilon}$, the proposed scheme is considered suboptimal, as there exists an alternative coding scheme that supports a higher rate than $R^\star$ without utilizing side information. Conversely, if $R^\star \ge R^\star_{\text{no-SI},\epsilon}$, the proposed scheme effectively leverages the available side information to outperform the no-side-information baseline. The following finite-blocklength regimes are considered to evaluate the proposed coding scheme.

1) Finite-blocklength Analysis for Binary Modulation: When the binary modulation $M = 2$ is adopted, the minimum achievable error probability can be determined, enabling us to evaluate whether the proposed scheme provides a reliability gain over the baseline.

2) Large-Blocklength Comparison: When the blocklength N is sufficiently large, the framework of Polyanskiy *et al.* [8] can be used to further identify the specific blocklength regimes where the coding scheme provides performance gains.

Theorem 1. (Normal approximation) [8, Theorem 54] For an AWGN channel with SNR ρ, the maximum achievable rate $R^\star(N, \epsilon)$ with fixed ρ and ϵ, satisfies the following expansion

$$R^\star(N, \epsilon) = C - \sqrt{\frac{2V}{N}}\,\mathrm{erfc}^{-1}(2\epsilon) + O\left(\frac{\log N}{N}\right), \tag{2}$$

where $C = \frac{1}{2}\log(1+\rho)$ is the channel capacity and $V = \frac{\rho(\rho+2)}{2(1+\rho)^2}(\log e)^2$ is the channel dispersion [8, Definition 1], which characterizes the stochastic variability of the channel.

Polyanskiy's normal approximation [8] provides an estimation of $R^\star_{\text{no-SI},\epsilon}$ as $\hat{R}^\star_{\text{no-SI},\epsilon} = C - \sqrt{\frac{2V}{N}}\,\mathrm{erfc}^{-1}(2\epsilon)$, where C and V share the same definitions as in Theorem 1, with the channel SNR $\rho = \frac{P}{\sigma^2+\sigma_s^2}$. However, $\hat{R}^\star_{\text{no-SI},\epsilon}$ is an asymptotic expansion, and it generally differs from the exact finite-blocklength rate $R^\star_{\text{no-SI},\epsilon}$. We characterize the gap by defining the approximation error $\Delta = R^\star_{\text{no-SI},\epsilon} - \hat{R}^\star_{\text{no-SI},\epsilon}$, which captures the higher-order terms neglected in the normal approximation and is dominated by terms of order $O\left(\frac{\log(N)}{N}\right)$. Accordingly, for a fixed, sufficiently large N and a target error probability ϵ, there exists a constant $\kappa > 0$ such that the exact finite-blocklength rate is bounded as

$$\hat{R}^\star_{\text{no-SI},\epsilon} - \frac{\kappa\log(N)}{N} \le R^\star_{\text{no-SI},\epsilon} \le \hat{R}^\star_{\text{no-SI},\epsilon} + \frac{\kappa\log(N)}{N}. \tag{3}$$

This bound accounts for the residual gap between the normal approximation and the exact finite-blocklength rate, providing a reliable benchmark for evaluating the proposed scheme.

III. Main Result

This paper first proposes the feedforward control-based coding framework in Section IV-A and formulates the linear coding scheme design within this framework in Section IV-B. Using the resulting closed-form expressions, Sections IV-C and IV-D analyze the post-processing signal-to-interference-plus-noise ratio (SINR) η and the probability of error for the proposed linear scheme respectively. These results establish

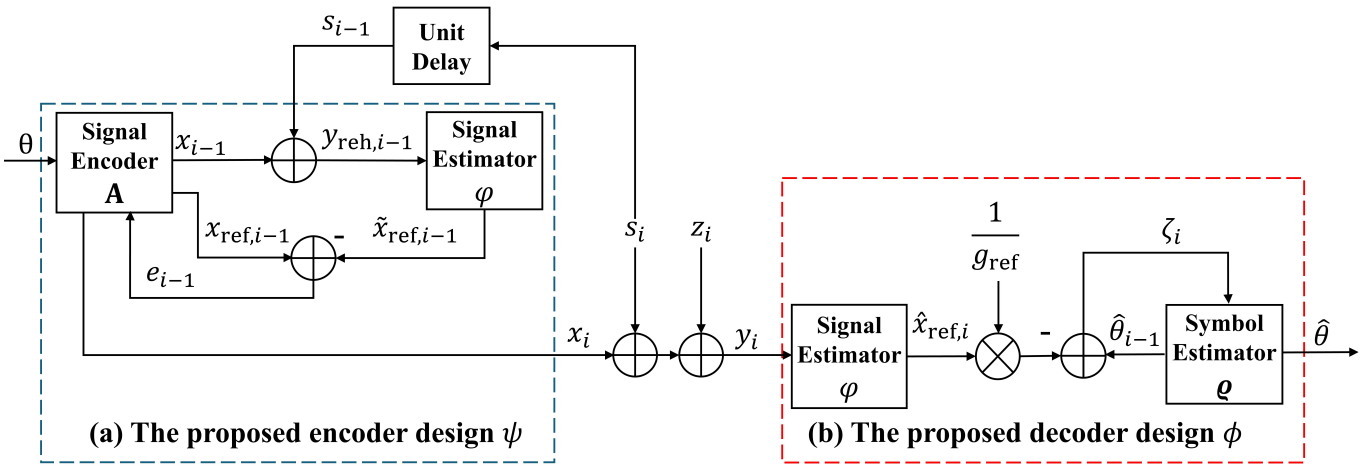


Fig. 2: The proposed coding framework for the strictly-causally known interference channel at channel use i. Following the ILC architecture, the same Signal Estimator φ serves as the robust filter at the encoder ψ and decoder ϕ, while the Signal Encoder parameterized by $\mathbf{A}$ in encoder ψ and the Symbol Estimator parameterized by $\boldsymbol{\varrho}$ in decoder ϕ serve as learning functions.

the following main two theorems characterizing the finite-blocklength achievable rate of the proposed scheme.

Theorem 2. For a modulation order of $M = 2$, it is defined the error probabilities as $\epsilon_0 = \frac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{\eta}{2}}\right)$ and $\epsilon_{\text{no-SI}} = \frac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{NP}{2(\sigma_s^2+\sigma^2)}}\right)$. Then, for every target error probability $\epsilon \in [\epsilon_0, \epsilon_{\text{no-SI}})$ and any blocklength $N > 1$, the proposed linear scheme achieves rate $\frac{1}{N}$, whereas any scheme without side information is limited to $R^{\star}_{\text{no-SI},\epsilon} = 0$.

Proof. The proof is contained in Appendices B of [19]. ■

Theorem 2 demonstrates the performance gain of the proposed coding scheme in Section IV for binary modulation $M = 2$. The scheme successfully supports a positive transmission rate while achieving a lower target error probability, a performance level that is fundamentally unattainable by any scheme lacking access to interference side information.

Theorem 3. Let $k = \frac{\sigma_s^2}{\sigma^2}$. Define $\rho = \frac{P}{\sigma^2+\sigma_s^2} > 0$ and κ as a constant. There exists a threshold $N_0 \in \mathbb{N}$ such that for any fixed $N > N_0$, there exists $k_0 > 1$ such that for all $k > k_0$, the achievable rate of the coding scheme satisfies $R^{\star} > R^{\star}_{\text{no-SI},\epsilon}$ under a fixed target error probability ϵ.

Proof. The proof is contained in Appendices B of [19]. ■

The simulation results in Appendix E of [19] further compares $R^{\star}$ with the converse bound of $R^{\star}_{\text{no-SI},\epsilon}$ [20]. This result in conjunction with Theorem 3 demonstrates that the proposed scheme can support a higher transmission rate to achieve a sufficiently small target error probability compared to the baseline scenario, where the encoder has no access to interference side information.

IV. Feedforward-Control-Based Coding Framework and Design

A. Proposed Coding Framework

Fig. 2 shows the ILC architecture of the proposed coding framework. A reference signal $x_{\text{ref},i}$ in Fig. 2 (a) is defined as the target system output without any disturbance. Accordingly, this paper considers a constant reference signal across the blocklength N, given by

$$x_{\text{ref},i} = g_{\text{ref}}\theta, \quad i = 1, \ldots, N. \tag{4}$$

Since $\mathbb{E}[\theta^2] = 1$, we choose $g_{\text{ref}}^2 < P$ to reserve power for later updates.

The proposed encoder ψ design is shown in Fig. 2 (a). The Signal Encoder serves as a *learning function* to generate and update the transmit signal $\mathbf{x} = [x_1, \cdots, x_N]^T$. It is initialized as $x_1 = x_{\text{ref},1} = g_{\text{ref}}\theta$. For $i \geq 2$, the Signal Encoder recursively exploits the available interference side information to construct the rehearsal observation as $y_{\text{reh},i-1} = x_{i-1} + s_{i-1}$. The Signal Estimator serves as a *robust filter* in Fig. 2 (a) to estimate the reference signal from the available causal rehearsal observations as $\tilde{x}_{\text{ref},i-1} = \varphi\left(y_{\text{reh},1}, \ldots, y_{\text{reh},i-1}\right)$, where $\varphi(\cdot)$ denotes the signal estimation function. The mismatch between the reference signal $x_{\text{ref},i-1}$ and its rehearsal estimation $\tilde{x}_{\text{ref},i-1}$ is defined as the rehearsal error $e_{i-1} = x_{\text{ref},i-1} - \tilde{x}_{\text{ref},i-1}$. The Signal Encoder then uses the last step rehearsal error to generate the transmit signal for the next channel use as $x_i = x_{i-1} + \alpha_{i-1}e_{i-1}$, where $\{\alpha_1, \cdots, \alpha_{N-1}\}$ denotes a set of encoding learning gains applied to the rehearsal errors. After accumulating the updates over all previous iterations, the transmit signal is expressed as $x_i = x_{\text{ref},1} + \sum_{j=1}^{i-1}\alpha_j\left(x_{\text{ref},j} - \hat{x}_{\text{ref},j}\right) = g_{\text{ref}}\theta + \sum_{j=1}^{i-1}\alpha_j e_j$. In matrix form, the encoder in Fig. 2 (a) is compactly expressed as

$$\mathbf{x} = g_{\text{ref}}\theta\mathbf{1} + \mathbf{A}\mathbf{e}, \tag{5}$$

where $\mathbf{1} \in \mathbb{R}^N$ denotes the all-one column vector, i.e., $\mathbf{1} = [1, \cdots, 1]^T$. $\mathbf{e} = [e_1, \ldots, e_N]^T$ is the rehearsal error vector. The encoding learning matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$ is a lower-triangular matrix with zero diagonal entries, where $a_{i,j}$ is the ith row and jth column entry of matrix $\mathbf{A}$ with $a_{i,j} = \begin{cases} \alpha_j, & j < i, \\ 0, & j \geq i, \end{cases}$ $i, j = 1, \ldots, N$. Since $\mathbf{A}$ is strictly lower triangular,e_N can be viewed as an arbitrary constant that does not affect the update of $\mathbf{x}$. The term $\mathbf{Ae}$ denotes the pre-compensating signal generated at the encoder.

The proposed decoder ϕ design is shown in Fig. 2 (b). The decoder successively updates its estimate of the symbol θ from y_i. As seen from Fig. 2 (b), the decoder employs the same signal estimation function φ as the encoder in Fig. 2 (a) to filter out the disturbances and obtain an estimation of the reference signal $\hat{x}_{\mathrm{ref},i} = \varphi(y_1, \ldots, y_i)$. The Symbol Estimator, parameterized by $\boldsymbol{\varrho} \in \mathbb{R}^N$, serves as the decoder-side *learning function* that iteratively estimates the transmitted symbol based on the mismatch between its previous estimation $\hat{\theta}_{i-1}$ and the estimated reference signal $\hat{x}_{\mathrm{ref},i}$. Specifically, this mismatch is defined as the estimation error $\zeta_i = \hat{\theta}_{i-1} - \frac{1}{g_{\mathrm{ref}}}\hat{x}_{\mathrm{ref},i}$, where $\zeta_1 = -\frac{1}{g_{\mathrm{ref}}}\hat{x}_{\mathrm{ref},1}$. The symbol estimate is iteratively updated as $\hat{\theta}_i = \hat{\theta}_{i-1} + \beta_i\zeta_i$, where $\{\beta_1, \ldots, \beta_N\}$ is a set of decoding learning gains. The estimated symbol $\hat{\theta}$ is $\hat{\theta} = \hat{\theta}_N = \sum_{i=1}^N \beta_i\zeta_i$. In vector form, the decoder output is

$$\hat{\theta} = \boldsymbol{\varrho}^T\boldsymbol{\zeta}, \tag{6}$$

where the decoding learning vector is $\boldsymbol{\varrho} = [\beta_1, \ldots, \beta_N]^T$, and the estimation error vector is $\boldsymbol{\zeta} = [\zeta_1, \ldots, \zeta_N]^T$. Following this framework, in what follows, a linear coding scheme is constructed.

B. Encoder and Decoder Design

The identical Signal Estimator φ at the encoder and decoder is parameterized by a linear filter denoted by a lower-triangular matrix $\mathbf{F}_{\mathrm{est}} \in \mathbb{R}^{N\times N}$, which estimates the rehearsal signal based on, respectively, the rehearsal observation $\mathbf{y}_{\mathrm{reh}} = [y_{\mathrm{reh},1}, \cdots, y_{\mathrm{reh},N}]^T$ at the encoder and the received signal $\mathbf{y} = [y_1, \cdots, y_N]^T$ at the decoder as

$$\tilde{\mathbf{x}}_{\mathrm{ref}} = \mathbf{F}_{\mathrm{est}}(\mathbf{x} + \mathbf{s}), \tag{7}$$

$$\hat{\mathbf{x}}_{\mathrm{ref}} = \mathbf{F}_{\mathrm{est}}\mathbf{y}, \tag{8}$$

where $\mathbf{s} = [s_1, \cdots, s_N]^T$. Then the rehearsal error $\mathbf{e}$ and the estimation error $\boldsymbol{\zeta}$ are expressed as

$$\mathbf{e} = g_{\mathrm{ref}}\theta\mathbf{1} - \mathbf{F}_{\mathrm{est}}(\mathbf{x} + \mathbf{s}), \tag{9}$$

$$\boldsymbol{\zeta} = \frac{1}{g_{\mathrm{ref}}}(\mathbf{B} - \mathbf{I})^{-1}\mathbf{F}_{\mathrm{est}}\mathbf{y}, \tag{10}$$

where $\mathbf{B} \in \mathbb{R}^{N\times N}$ is a strictly lower-triangular matrix constructed from the decoding learning gains $\{\beta_i\}_{i=1}^N$, with each element given by $b_{i,j} = \begin{cases} \beta_j, & j < i, \\ 0, & j \geq i. \end{cases}$ From (5) and (9), the resulting transmitted signal $\mathbf{x}$ can be rewritten as

$$\mathbf{x} = \mathbf{g}\theta + \mathbf{F}\mathbf{s}, \tag{11}$$

where $\mathbf{g} \in \mathbb{R}^N$ is given by $\mathbf{g} = g_{\mathrm{ref}}\left(\mathbf{I} + \mathbf{A}\mathbf{F}_{\mathrm{est}}\right)^{-1}(\mathbf{I} + \mathbf{A})\,\mathbf{1}$, and a lower-triangular matrix $\mathbf{F} \in \mathbb{R}^{N\times N}$ with zero diagonal entries is given by $\mathbf{F} = -\left(\mathbf{I} + \mathbf{A}\mathbf{F}_{\mathrm{est}}\right)^{-1}\mathbf{A}\mathbf{F}_{\mathrm{est}}$. The channel output is then

$$\mathbf{y} = \mathbf{x} + \mathbf{s} + \mathbf{z} = \mathbf{g}\theta + (\mathbf{I} + \mathbf{F})\mathbf{s} + \mathbf{z}. \tag{12}$$

From (6) and (10), the decoder is equivalent to a linear combiner $\mathbf{w} \in \mathbb{R}^{N\times 1}$ to estimate the θ from $\mathbf{y}$,

$$\hat{\theta} = \mathbf{w}^T\mathbf{y} = \mathbf{w}^T\mathbf{g}\theta + \mathbf{w}^T\big((\mathbf{F} + \mathbf{I})\mathbf{s} + \mathbf{z}\big), \tag{13}$$

where $\mathbf{w} = \frac{1}{g_{\mathrm{ref}}}\mathbf{F}_{\mathrm{est}}^T\left[(\mathbf{B} - \mathbf{I})^{-1}\right]^T\boldsymbol{\varrho}$. The expression in (13) is the final system output, which consists of two components: the message component $\mathbf{w}^T\mathbf{g}\theta$ and the disturbance-induced component

$$\hat{\theta}_d = \mathbf{w}^T\big((\mathbf{F} + \mathbf{I})\mathbf{s} + \mathbf{z}\big). \tag{14}$$

With the message symbol θ constant over the N channel uses, the parameter designs of $\mathbf{g}$, $\mathbf{F}$, and $\mathbf{w}$ reduce to a disturbance-attenuation problem [21], [22], targeting $\hat{\theta}_d$ suppression in (14). An objective function for this problem is considered as the relative sensitivity [22], in which a large sensitivity indicates that small changes in $\mathbf{s}$ or $\mathbf{z} = [z_1, \cdots, z_N]^T$ can cause large variations in $\hat{\theta}$ and vice versa.

Definition 1 (Relative Sensitivity)**.** [23, (1)] Let $\mathcal{T}$ denote a system response with respect to a parameter v. The relative sensitivity is defined as

$$S_v^{\mathcal{T}} \triangleq \frac{v}{\mathcal{T}}\frac{\partial\mathcal{T}}{\partial v}. \tag{15}$$

Based on (15), the sensitivity is the relationship between the power of the system output, $\mathbb{E}[\hat{\theta}^2]$, and the powers of the disturbance-induced component, $\mathbb{E}[\hat{\theta}_d^2]$. Then according to (1), the channel model leads to the following sensitivity-based objective function:

$$\begin{aligned}\mathcal{L} &= \frac{\mathbb{E}[\hat{\theta}_d^2]}{\mathbb{E}[\hat{\theta}^2]}\frac{\partial\mathbb{E}[\hat{\theta}^2]}{\partial\mathbb{E}[\hat{\theta}_d^2]} \\ &= \frac{\mathbb{E}\Big[\big|\mathbf{w}^T\big(\mathbf{F} + \mathbf{I})\mathbf{s} + \mathbf{z}\big)\big|^2\Big]}{\mathbb{E}[|\mathbf{w}^T\mathbf{g}\theta|^2] + \mathbb{E}\big[\big|\mathbf{w}^T\big((\mathbf{F} + \mathbf{I})\mathbf{s}\big|^2\big] + \mathbb{E}\left[|\mathbf{w}^T\mathbf{z}|^2\right]}.\end{aligned} \tag{16}$$

Under the transmit power constraint the following sensitivity minimization problem is formulated:

$$\begin{aligned}(\mathcal{K}): \quad &\min_{\mathbf{g},\mathbf{F},\mathbf{w}} \mathcal{L} = \frac{\mathbb{E}[\hat{\theta}_d^2]}{\mathbb{E}[\hat{\theta}^2]}\frac{\partial\mathbb{E}[\hat{\theta}^2]}{\partial\mathbb{E}[\hat{\theta}_d^2]} \\ &\text{subject to} \quad \sum_{i=1}^N \mathbb{E}[x_i^2] = \|\mathbf{g}\|_2^2 + \sigma_s^2\,\mathrm{tr}(\mathbf{F}\mathbf{F}^T) \leq NP.\end{aligned} \tag{17}$$

Lemma 1. The solution to the power-constrained sensitivity minimization problem in $(\mathcal{K})$ is given by

$$f_{i+j,i} = -(\delta - 1)\delta^{-j/2}, \quad j = 1, \ldots, N - i, \tag{18}$$

$$g_i = \sqrt{(\sigma_s^2 + \sigma^2\delta^{N-1})(\delta - 1)\delta^{-(i-1)}}, \tag{19}$$

$$\hat{\theta} = \mathbf{w}^T\mathbf{y} = \frac{\mathbf{g}^T\mathbf{C}^{-1}\mathbf{y}}{\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g}}, \quad \mathbf{w} = \frac{\mathbf{C}^{-1}\mathbf{g}}{\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g}}. \tag{20}$$

where $f_{i+j,i}$ denotes the $(i+j)$th row and ith column entry of $\mathbf{F}$, $\mathbf{C} = \sigma_s^2(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T + \sigma^2\mathbf{I}$ is the interference-plus-noise

covariance matrix, and $\delta = (1+\eta)^{1/N}$, where η is determined as the root of

$$\sigma^2\eta - \sigma_s^2 N + \sigma_s^2 N(1+\eta)^{1/N} = NP. \tag{21}$$

Proof. The proof is relegated to Appendix A in [19]. ■

C. From Sensitivity to SINR

By (20), it is obtained that $\mathbf{w}^T\mathbf{g} = 1$. Given $\mathbf{F}$ in (18) and $\mathbf{g}$ in (19), the optimal sensitivity is given by $\mathcal{L} = \frac{1}{1+\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g}} = \frac{1}{1+\eta}$. The equality in (21) leads to $\frac{\eta}{N} = \frac{P}{\sigma^2} - \frac{\sigma_s^2}{\sigma^2}\left((1+\eta)^{1/N} - 1\right) \leq \frac{P}{\sigma^2}$. On the other hand, $NP \leq \sigma^2\eta - \sigma_s^2 N + \sigma_s^2 N\left(1+\frac{\eta}{N}\right) = (\sigma^2+\sigma_s^2)\eta$, where the inequality follows from the fact that $(1+\eta)^{1/N} \leq (1+\eta/N)$ for $N \geq 1$. These together yield

$$N\left(\frac{P}{\sigma^2+\sigma_s^2}\right) \leq \eta \leq N\frac{P}{\sigma^2}. \tag{22}$$

Regarding communication performance, the SINR quantifies interference mitigation. Under the proposed linear scheme, the SINR evaluates to $\text{SINR} = \frac{\mathbb{E}\left[\left|\mathbf{w}^T\mathbf{g}\theta\right|^2\right]}{\mathbb{E}\left[\left|\mathbf{w}^T\left((\mathbf{F}+\mathbf{I})\mathbf{s}+\mathbf{z}\right)\right|^2\right]} = \mathbf{g}\mathbf{C}^{-1}\mathbf{g} = \eta$. Therefore, minimizing our sensitivity objective function $\mathcal{L}$ is as maximizing the SINR. The lower bound in (22) is achieved when $N = 1$. The upper bound in (22) becomes tighter as N grows, demonstrating that the proposed linear scheme effectively suppresses interference as $N \to \infty$.

D. Probability of Error Analysis

The likelihood $p(\hat{\theta}|\theta)$ is conditionally Gaussian distributed with mean $E[\hat{\theta}|\theta] = \theta$ and variance $\mathbb{E}\left[(\hat{\theta}-\theta)^2 \mid \theta\right] = \frac{1}{\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g}} = \frac{1}{\eta}$, yielding $p\left(\hat{\theta} \mid \theta\right) = \sqrt{\frac{\eta}{2\pi}}\exp\left(-\frac{\eta(\hat{\theta}-\theta)^2}{2}\right)$. Since the transmit symbol is equiprobable, an error occurs if there exists a symbol $\theta' \neq \theta$, $\theta', \theta \in \mathcal{C}$ such that $|\theta' - \hat{\theta}| \leq |\theta - \hat{\theta}|$. For M-ary pulse amplitude modulation (PAM), the distance between adjacent symbols is given by $d = 2\sqrt{\frac{3}{M^2-1}}$. Following the standard derivation for the PAM probability of error [24], the maximal probability of error $P_{e,\max}$ is expressed as

$$P_{e,\max} = \begin{cases} \frac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{\eta}{2}}\right), & M = 2, \\ \operatorname{erfc}\left(\sqrt{\frac{3\eta}{2(M^2-1)}}\right), & M \geq 3. \end{cases} \tag{23}$$

where $\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}}\int_x^\infty e^{-t^2}\,dt$. For a fixed M, the proposed scheme ensures that the maximal error probability in (23) decays exponentially with the blocklength N.

Lemma 2. For the linear scheme with a fixed target error probability $\epsilon \in \left[\operatorname{erfc}\left(\sqrt{\frac{1}{2}\eta}\right), 1\right)$, there exists some $\nu \in (0, N/3]$ such that the rate $R^\star = \frac{\log(\lfloor\sqrt{1+\frac{N}{\nu}}\rfloor)}{N}$ is achievable.

Proof. From (22), for any $\tau \in \left[\frac{P}{\sigma^2+\sigma_s^2}, \frac{P}{\sigma^2}\right]$, we get $\eta = N\tau$. For an achievable $R^\star$, there exists $\nu > 0$ such that $R^\star = \frac{\log(\lfloor\sqrt{1+\frac{N}{\nu}}\rfloor)}{N}$, in which the error probability is $P_{e,\max} < \operatorname{erfc}\left(\sqrt{\frac{3}{2}\nu\tau}\right)$. Since $M \geq 2$, ν should satisfy $\nu \leq \frac{N}{3}$, implying $\epsilon \geq \operatorname{erfc}\left(\sqrt{\frac{\eta}{2}}\right)$. Since $0 \leq \nu \leq \frac{N}{3}$, the proposed linear scheme achieves target error probability $\epsilon \in \left[\operatorname{erfc}\left(\sqrt{\frac{\eta}{2}}\right), 1\right)$ with rate $R^\star = \frac{\log(\lfloor\sqrt{1+\frac{N}{\nu}}\rfloor)}{N}$. This completes the proof. ■

Remark 1. Lemma 2 provides a finite-blocklength characterization of the proposed linear coding scheme. It shows that the proposed coding scheme can support a non-zero rate while maintaining a target error probability $\epsilon \in [\operatorname{erfc}\left(\sqrt{\frac{1}{2}\eta}\right), 1)$. This motivates the selection of appropriate values for the transmission rate $R^\star$, message size M, error probability ϵ, and blocklength N to ensure the effective operation of the scheme. Moreover, Lemma 2 also indicates that the interference power σ_s^2 and the noise power σ^2 are the primary factors affecting reliability performance.

The error probability analysis in (23), together with Lemma 2, yields Theorems 2 and 3. Due to space limitations, their proofs and the converse comparison results are delayed to the Appendix in [19] with the locations specified in Section III.

V. Conclusion

This paper proposed a feedforward-control based coding framework and a finite-blocklength linear code for the strictly-causally known interference channel.The analysis demonstrates finite-blocklength achievable rate gains of the proposed scheme in interference-dominated regimes. It is noteworthy that the proposed linear scheme is one realization of the proposed framework, alternative coding constructions using ILC, particularly for the noise-dominated regime, remain an open problem. A promising future work is to develop a hybrid coding scheme that augments the existing linear code with residual learning [25]. Joint encoder-decoder training enables the encoder to learn the true encoder–decoder mismatch during training and further improve robustness for unknown channel states.

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APPENDIX A
SOLUTION TO PROBLEM $(\mathcal{K})$

A. Linear Decoder Design

The objective function $\mathcal{L}$ in (16) is simplified as

$$\mathcal{L}=\frac{\mathbf{w}^T\mathbf{C}\mathbf{w}}{|\mathbf{w}^T\mathbf{g}|^2+\mathbf{w}^T\mathbf{C}\mathbf{w}}, \tag{24}$$

where $\mathbf{C}=\sigma_s^2(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T+\sigma^2\mathbf{I}$ is the interference-plus-noise covariance matrix. The decoding vector $\mathbf{w}$ is first determined while fixing $\mathbf{g}$ and $\mathbf{F}$. By the Cauchy–Schwarz inequality, the following holds $|\mathbf{w}^T\mathbf{g}|^2\le(\mathbf{w}^T\mathbf{C}\mathbf{w})(\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g})$. and $\mathcal{L}$ is lower bounded as

$$\mathcal{L}\ge\frac{\mathbf{w}^T\mathbf{C}\mathbf{w}}{\mathbf{w}^T\mathbf{C}\mathbf{w}(\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g})+\mathbf{w}^T\mathbf{C}\mathbf{w}}=\frac{1}{1+\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g}}. \tag{25}$$

The lower bound is achieved by choosing an unbiased decoder weight,

$$\mathbf{w}=\frac{\mathbf{C}^{-1}\mathbf{g}}{\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g}}.$$

Thus, by setting $\eta=\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g}$ in (25), the sensitivity minimization problem $(\mathcal{K})$ is transformed to maximizing η under the power constraint. From eigenvalue decomposition, η is upper-bounded by $\eta\le\frac{\mathbf{g}^T\mathbf{g}}{\lambda_{\min}(\mathbf{C})}$, where the minimum eigenvalue of $\mathbf{C}$ is denoted as $\lambda_{\min}(\mathbf{C})$ and the upper bound of η is achieved when $\mathbf{g}$ is the eigenvector associated with $\lambda_{\min}(\mathbf{C})$, such that $\mathbf{C}\mathbf{g}=\lambda_{\min}(\mathbf{C})\mathbf{g}$. Then, the problem $(\mathcal{K})$ is reformulated as

$$\begin{aligned}(\mathcal{J}):\quad \max_{\mathbf{g},\mathbf{F}}\quad & \eta=\frac{\mathbf{g}^T\mathbf{g}}{\lambda_{\min}(\mathbf{C})}\\ \text{subject to}\quad & \mathbf{C}\mathbf{g}=\lambda_{\min}(\mathbf{C})\mathbf{g},\\ & \|\mathbf{g}\|_2^2+\sigma_s^2\operatorname{tr}(\mathbf{F}\mathbf{F}^T)\le NP.\end{aligned} \tag{26}$$

B. Linear Encoder Design

For a fixed $\mathbf{F}$, $\lambda_{\min}(\mathbf{C})$ is independent of $\mathbf{g}$. η is monotonic with respect to $\mathbf{g}^T\mathbf{g}$. Hence, the power constraint in (26) is active at the optimum of $\mathcal{J}$, which yields $\mathbf{g}^T\mathbf{g}=NP-\sigma_s^2\operatorname{tr}(\mathbf{F}\mathbf{F}^T)$, and consequently,

$$\eta=\frac{NP-\sigma_s^2\operatorname{tr}(\mathbf{F}\mathbf{F}^T)}{\lambda_{\min}(\mathbf{C})}. \tag{27}$$

Therefore, the maximization of η solely depends on $\mathbf{F}$. $\lambda_1\le\cdots\le\lambda_N$ are denoted as the eigenvalues of $(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T$, with the minimum eigenvalue denoted by $\lambda_1=\lambda$. Since the matrix $\mathbf{F}$ is a lower-triangular matrix with zero diagonal entries, it is obtained that $\det((\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T)=1$ and $\Pi_{i=2}^N\lambda_i=\frac{1}{\lambda}$. By the arithmetic mean and geometric mean inequality, and setting $\delta=\lambda^{-\frac{1}{N-1}}$.

$$\operatorname{tr}\left((\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T\right)=\sum_{i=1}^N\lambda_i\ge\delta^{-(N-1)}+(N-1)\delta, \tag{28}$$

$$\operatorname{tr}(\mathbf{F}\mathbf{F}^T)\ge\delta^{-(N-1)}+(N-1)\delta-N. \tag{29}$$

The inequalities in (28) and (29) holds as the equalities when

$$\lambda_i=\begin{cases}\delta^{-(N-1)}, & i=1,\\ \delta, & i=2,\dots,N.\end{cases} \tag{30}$$

The minimum eigenvalue of $\mathbf{C}$ is then given by

$$\lambda_{\min}(\mathbf{C})=\sigma_s^2\delta^{-(N-1)}+\sigma^2. \tag{31}$$

Thus, η is upper-bounded by

$$\eta\le D(\delta)\triangleq\frac{NP-\sigma_s^2\left[\delta^{-(N-1)}+(N-1)\delta-N\right]}{\sigma^2+\sigma_s^2\delta^{-(N-1)}}. \tag{32}$$

Differentiating $D(\delta)$ with regards to δ gives

$$\frac{dD(\delta)}{d\delta}=\frac{(N-1)\sigma_s^2\delta^{N-2}\left[NP+N\sigma_s^2+\sigma^2-N\sigma_s^2\delta-\sigma^2\delta^N\right]}{\left(\sigma^2\delta^{N-1}+\sigma_s^2\right)^2}. \tag{33}$$

Hence, as δ grows, $D(\delta)$ first increases and then decreases, and thus admits a unique global maximum for $\delta>1$, which satisfies

$$\sigma^2(\delta^N-1)-N\sigma_s^2+N\sigma_s^2\delta=NP. \tag{34}$$

From (34), the optimal value of η in (32) is given by

$$\delta=(1+\eta)^{\frac{1}{N}}. \tag{35}$$

The optimal $\mathbf{F}$ is then of interest. Based on (30), $\mathbf{u}=[u_1,\dots,u_N]^T$ is defined as a unit eigenvector associated with $\lambda_1=\delta^{-(N-1)}$. Since the remaining $N-1$ eigenvalues are all equal to δ, the eigenvalue decomposition of $(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T$ gives

$$(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T=\delta\mathbf{I}+\left(\delta^{-(N-1)}-\delta\right)\mathbf{u}\mathbf{u}^T. \tag{36}$$

Since $\mathbf{F}$ is a lower-triangular matrix with zero diagonal entries, every leading principal minor of $(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T$ equals one. Therefore, $\delta^i\left[1+(\delta^{-N}-1)\sum_{j=1}^i u_j^2\right]=1,\quad i=1,\dots,N$, where the matrix determinant lemma for the rank-one update has been applied to (36). Hence, $\sum_{j=1}^i u_j^2=\frac{1-\delta^{-i}}{1-\delta^{-N}}$, and consequently $u_i^2=\frac{(\delta-1)\delta^{N-i}}{\delta^N-1}$. For $u_i>0$,

$$u_i=\sqrt{\frac{(\delta-1)\delta^{N-i}}{\delta^N-1}}. \tag{37}$$

Substituting (37) into (36) yields $\left[(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T\right]_{i+j,i}=-(\delta-1)\delta^{-\frac{2i+j-2}{2}},\quad j=1,\dots,N-i$, where $[(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T]_{i+j,j}$ denotes the entry in the $(i+j)$th row and jth column of $(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T$. By applying Cholesky decomposition to $(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T$, the corresponding entries of $\mathbf{F}$ are given by

$$f_{i+j,i}=-(\delta-1)\delta^{-j/2},\qquad j=1,\dots,N-i. \tag{38}$$

Since $\mathbf{g}$ is also the eigenvector corresponding to $\lambda_{\min}(\mathbf{C})$, it holds $\mathbf{g}=c\mathbf{u}$, where $c\in\mathbb{R}$ is a constant. The value of c is determined by the equality $\mathbf{g}^T\mathbf{g}=NP-\sigma_s^2\operatorname{tr}(\mathbf{F}\mathbf{F}^T)$, as $c=\sqrt{NP-\sigma_s^2\operatorname{tr}(\mathbf{F}\mathbf{F}^T)}=\sqrt{\left(\sigma^2+\sigma_s^2\delta^{-(N-1)}\right)(\delta^N-1)}$. Then the ith entry of $\mathbf{g}$ is given by

$$g_i=\sqrt{(\sigma_s^2+\sigma^2\delta^{N-1})(\delta-1)\delta^{-(i-1)}},\quad 1,\cdots,N. \tag{39}$$

APPENDIX B
PROOF OF THEOREM 2

First, we examine the AWGN interference channel without encoder side information. For $M=2$, the message set contains two equiprobable symbols $\theta_1,\theta_2\in\mathcal{C}$, encoded as equiprobable code vectors $\mathbf{x}_1,\mathbf{x}_2\in\mathbb{R}^N$. Bounded by the

power constraint $\mathbf{x}_1^T\mathbf{x}_1 + \mathbf{x}_2^T\mathbf{x}_2 \leq 2NP$, the achievable error probability [24] is

$$P_{e,\text{no-SI}} = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{d_{\text{no-SI}}^2}{8(\sigma_s^2+\sigma^2)}}\right), \tag{40}$$

where $d_{\text{no-SI}} = \|\mathbf{x}_1 - \mathbf{x}_2\|$ is the Euclidean distance between the two codewords. Its square is bounded by $d_{\text{no-SI}}^2 = \|\mathbf{x}_1 - \mathbf{x}_2\|^2 = \|\mathbf{x}_1\|^2 + \|\mathbf{x}_2\|^2 - 2\mathbf{x}_1^T\mathbf{x}_2 \leq 2\|\mathbf{x}_1\|^2 + 2\|\mathbf{x}_2\|^2 \leq 4NP$. The maximum distance $d_{\text{no-SI}}$ is obtained when $\frac{\mathbf{x}_1^T\mathbf{x}_2}{\|\mathbf{x}_1\|\|\mathbf{x}_2\|} = -1$, meaning the signals are antipodal. The minimum achievable $P_{e,\text{no-SI}}$ is obtained as

$$P_{e,\text{no-SI,min}}^{\star} = \frac{1}{2}\,\text{erfc}\left(\sqrt{\frac{NP}{2(\sigma_s^2+\sigma^2)}}\right) = \epsilon_{\text{no-SI}}. \tag{41}$$

This indicates that any target error probability $\epsilon < \epsilon_{\text{no-SI}}$ is unattainable when $M = 2$. In other words, the achievable rate $R_{\text{no-SI},\epsilon}^{\star} = 0$ for any $\epsilon < \epsilon_{\text{ no-SI}}$.

When $M > 2$, we assume a no side information code achieves a target probability of error $\epsilon < \epsilon_{\text{no-SI}}$. For two code vectors $\mathbf{x}_A, \mathbf{x}_B \in \mathbb{R}^{\star}$, the average power constraint implies

$$\|\mathbf{x}_A - \mathbf{x}_B\|^2 \leq \frac{2MNP}{M-1} < 4NP. \tag{42}$$

Mapping every decoder output other than $\mathbf{x}_A$ to $\mathbf{x}_B$ yields a binary decoder with the error probability at most ϵ. However, its minimum error probability satisfies

$$P_{e,\text{no-SI,min}} \geq \frac{1}{2}\,\text{erfc}\left(\frac{\|\mathbf{x}_A - \mathbf{x}_B\|}{\sqrt{8(\sigma_s^2+\sigma^2)}}\right) > \epsilon_{\text{no-SI}} > \epsilon. \tag{43}$$

It is a contradiction. This means that any target error probability $\epsilon < \epsilon_{\text{no-SI}}$ is also unattainable when $M > 2$.

For the proposed scheme, applying $M = 2$ to (23) yields the achieved probability of error,

$$P_{e,\max} = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{1}{2}\eta}\right). \tag{44}$$

Leveraging the SINR bound in (22), $P_{e,\max}$ is bounded by $\frac{1}{2}\text{erfc}\left(\sqrt{\frac{NP}{2\sigma^2}}\right) \leq P_{e,\max} \leq \frac{1}{2}\text{erfc}\left(\sqrt{\frac{NP}{2(\sigma^2+\sigma_s^2)}}\right)$. By setting $P_{e,\max} = \epsilon$, it follows that for every $\epsilon \in [\epsilon_0, \epsilon_{\text{ no-SI}})$, the scheme achieves a strictly positive rate of $R^{\star} = \frac{1}{N}$, whereas $R_{\text{no-SI},\epsilon}^{\star} = 0$. This completes the proof.

Appendix C
Proof of Theorem 3

We fix the target probability of error $\epsilon < 1/2$, and set $\rho = \frac{P}{\sigma^2+\sigma_s^2}$ as a constant. According to the SINR bound in (22), there exists N_0 such that for $N > N_0$,

$$\epsilon \geq \text{erfc}\left(\sqrt{\frac{N\rho}{2}}\right) \geq \text{erfc}\left(\sqrt{\frac{\eta}{2}}\right). \tag{45}$$

When $k > 1$, the SINR η is characterized from (21) by $\eta = \left(1 + \frac{P}{\sigma_s^2} - \frac{\eta}{kN}\right)^N - 1$, which yields the following lower bound

$$\eta \geq \frac{k\left[\left(1+\frac{P}{\sigma_s^2}\right)^N - 1\right]}{k + \left(1+\frac{P}{\sigma_s^2}\right)^{N-1}}. \tag{46}$$

The detailed derivation of (46) is provided in Appendix D. By utilizing the error probability expression in (23), the linear coding scheme's achievable rate $R^{\star}$ for a given target error probability ϵ satisfies

$$\begin{aligned} R^{\star} &\geq \frac{1}{N}\log\left(\lfloor\sqrt{\frac{3\eta}{2(\text{erfc}^{-1}(\epsilon))^2}+1}\rfloor\right) \\ &\overset{(a)}{>} \frac{1}{2N}\log(\eta) - \frac{\log(\text{erfc}^{-1}(\epsilon))}{N} - \frac{1}{N} \\ &\overset{(b)}{\geq} \frac{1}{2}\log\left(1+\frac{P}{\sigma_s^2}\right) - \frac{1}{2N}\log\left(\frac{1+\frac{(1+\frac{P}{\sigma_s^2})^{N-1}}{k}}{1-(1+\frac{P}{\sigma_s^2})^{-N}}\right) \\ &\quad - \frac{\log(\text{erfc}^{-1}(\epsilon))}{N} - \frac{1}{N}, \end{aligned} \tag{47}$$

where (a) is due to the condition in (45) and the fact $\lfloor\ell\rfloor \geq \ell/2$ for all integers $\ell \geq 1$. (b) is due to (46) and $\log\left((1+\frac{P}{\sigma_s^2})^N - 1\right) = \log\left((1+\frac{P}{\sigma_s^2})^N\right) + \log\left(1 - (1+\frac{P}{\sigma_s^2})^{-N}\right)$. We denote $A(k) = \frac{1}{2}\log\left(\frac{1+\frac{\left(1+\rho\left(1+\frac{1}{k}\right)\right)^{N-1}}{k}}{1-\left(1+\rho\left(1+\frac{1}{k}\right)\right)^{-N}}\right)$, and $C = \frac{1}{2}\log(1+\frac{P}{\sigma_s^2+\sigma^2})$. The limit of $A(k)$ as $k \to \infty$ is given by

$$G(N) = \lim_{k\to\infty} A(k) = \frac{1}{2}\log\left(\frac{1}{1-(1+\rho)^{-N}}\right). \tag{48}$$

Since ρ is fixed as a positive constant, the case $\rho \to 0$ is excluded. Therefore, for any $\varepsilon' > 0$, there exists a k', such that for any $k > k' > 1$, we have $|A(k) - G(N)| \leq \varepsilon'$. We set $\varepsilon' = \frac{G(N)}{2}$ and then $A(k)$ is upper bounded by $A(k) \leq \frac{3G(N)}{2}$. Since $G(N)$ is decreasing with N, there exists N_1 such that for any fixed $N > N_1$, $A(k) < 2$ for $k > k'$, and from (47),

$$R^{\star} > C - \frac{3}{N} - \frac{\log(\text{erfc}^{-1}(\epsilon))}{N}. \tag{49}$$

From (3), since $\epsilon < \frac{1}{2}$, the baseline rate $R_{\text{no-SI},\epsilon}^{\star}$ satisfies

$$R_{\text{no-SI},\epsilon}^{\star} \leq C - \sqrt{\frac{2V}{N}}\text{erfc}^{-1}(2\epsilon) + \kappa\frac{\log N}{N}. \tag{50}$$

From (49) and (50), the gap between the two rates is lower bounded as $N(R^{\star} - R_{\text{no-SI},\epsilon}^{\star}) \geq \sqrt{2VN}\,\text{erfc}^{-1}(2\epsilon) - \kappa\log(N) - \log\left(\text{erfc}^{-1}(\epsilon)\right) - 3$. We now define a continuous function $B : \mathbb{R} \to \mathbb{R}$ such that $B(t)$ represents this lower bound for each $t \in \mathbb{R}$.

$$B(t) = \sqrt{2Vt}\text{erfc}^{-1}(2\epsilon) - \kappa\log(t) - \log\left(\text{erfc}^{-1}(\epsilon)\right) - 3. \tag{51}$$

If $B(N) > 0$, it guarantees $R^{\star} > R_{\text{no-SI},\epsilon}^{\star}$ when N is sufficiently large for $k > k'$. To analyze its behavior, we differentiate $B(t)$ with respect to t and set the derivative equal to zero: $\frac{dB(t)}{dt} = \frac{\text{erfc}^{-1}(2\epsilon)\sqrt{2V}}{2\sqrt{t}} - \frac{\kappa}{t\ln 2} = 0$. The stationary point is obtained by $t^{\star} = \frac{2\kappa^2}{V(\ln 2)^2\left(\text{erfc}^{-1}(2\epsilon)\right)^2}$. This implies that for $t > t^{\star}$, $B(N)$ is monotonically increasing. Consequently, there exists an $N_2 > t^{\star}$ such that for $N > N_2$, $B(N) > 0$. In conclusion, for a fixed blocklength $N > \max(N_0, N_1, N_2)$, there exists $k_0 > k'$ such that for any $k > k_0$, $R_{\text{no-SI},\epsilon}^{\star} < R^{\star}$. This completes the proof.

APPENDIX D
DERIVATIONS OF SINR FOR $\sigma_s^2 > \sigma^2$

For $k = \frac{\sigma_s^2}{\sigma^2} > 1$, we have

$$\begin{aligned}\eta + 1 &= \left(1 + \frac{P}{\sigma_s^2} - \frac{\eta}{kN}\right)^N \\ &= \left(1 + \frac{P}{\sigma_s^2}\right)^N \left(1 - \frac{\eta}{kN\left(1 + \frac{P}{\sigma_s^2}\right)}\right)^N \\ &\geq \left(1 + \frac{P}{\sigma_s^2}\right)^N \left(1 - \frac{\eta}{k\left(1 + \frac{P}{\sigma_s^2}\right)}\right),\end{aligned} \tag{52}$$

where the last inequality follows from Bernoulli's inequality, $(1-x)^N \geq 1 - Nx, x \in [0,1]$, with $x = \frac{\eta}{kN\left(1+\frac{P}{\sigma_s^2}\right)}$. Rearranging (52) yields $\eta \left(1 + \frac{\left(1+\frac{P}{\sigma_s^2}\right)^{N-1}}{k}\right) \geq \left(1 + \frac{P}{\sigma_s^2}\right)^N - 1$.

Therefore, $\eta \geq \frac{\left(1+\frac{P}{\sigma_s^2}\right)^N - 1}{1 + \frac{\left(1+\frac{P}{\sigma_s^2}\right)^{N-1}}{k}}$.

APPENDIX E
SIMULATION RESULTS ABOUT FINITE-BLOCKLENGTH ACHIEVABLE RATE

TABLE I: Parameter settings for finite-blocklength achievable rate comparison

ρ	N	ϵ
1	100	10^{-5}
10	100	10^{-5}
100	100	10^{-5}

The achievable rate of our linear coding scheme $R^\star$ is compared with the meta-converse bound and the normal approximation [8] of the baseline benchmark $R^\star_{\text{no-SI},\epsilon}$. The converse result is from [20, (36)] with real AWGN channel. Although it is derived under an average error probability constraint, this expression still provides a loose upper bound on $R^\star_{\text{no-SI},\epsilon}$, since the maximal error probability constraint is more stringent. The blocklength N, $\rho = \frac{P}{\sigma_s^2+\sigma^2}$, and the target block error rate (BLER) ϵ are fixed as a constant while varying the power allocation $\frac{\sigma_s^2}{\sigma_s^2+\sigma^2}$ as a ratio between the interference power over the total disturbance power. Table I summarizes the parameter settings used for comparison.

The simulation results are shown in Fig. 3. The results demonstrate that $R^\star$ increases as the fraction of interference power increases, whereas the baseline rate $R^\star_{\text{no-SI},\epsilon}$ remains constant since $\frac{P}{\sigma_s^2+\sigma^2}$ is fixed. When $\frac{\sigma_s^2}{\sigma_s^2+\sigma^2} \to 1$, $R^\star$ exceeds the converse bound for $R^\star_{\text{no-SI},\epsilon}$.

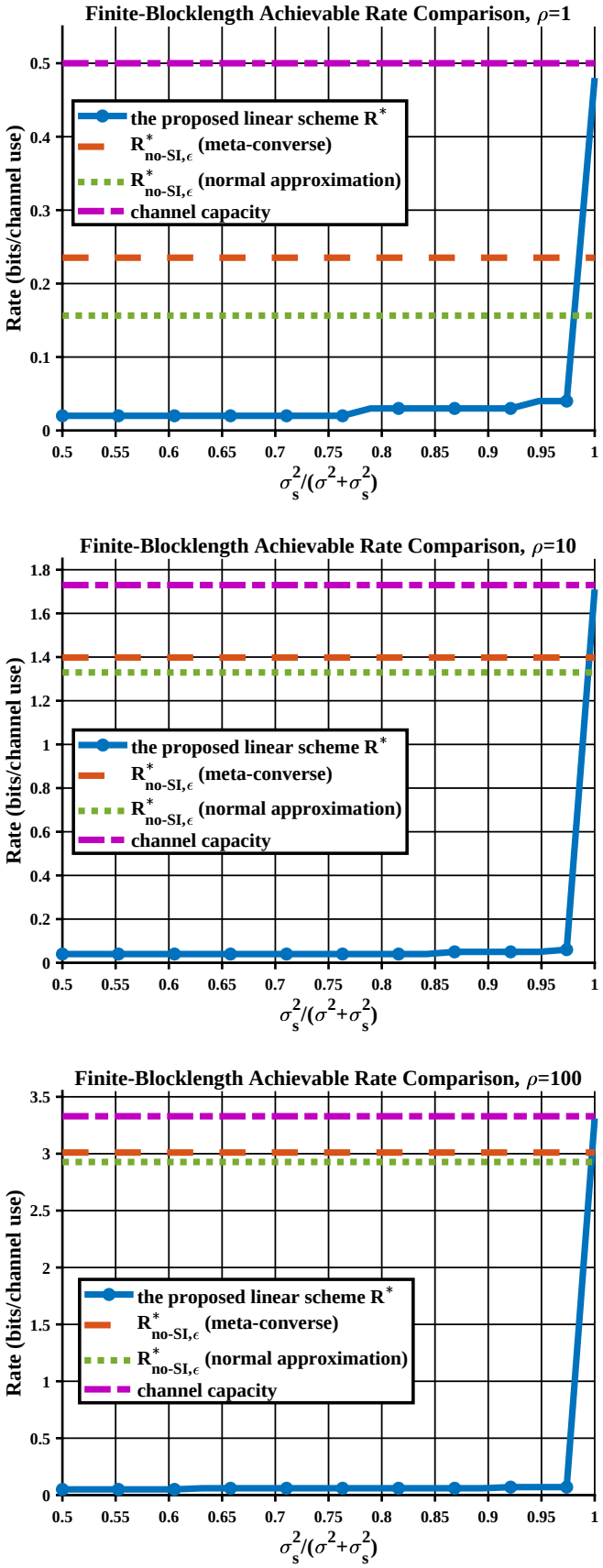


Fig. 3: Finite-blocklength achievable rates with different interference power allocations under a fixed $\rho = \frac{P}{\sigma_s^2+\sigma^2} \in \{1, 10, 100\}$, with $N = 100$ and target BLER $\epsilon = 10^{-5}$.